Sinclair ZX Spectrum 48K — Pinouts

Z80A · ULA · 4116 RAM · ROM · MULTI-RAIL PSU

DIP pinouts from the top, notch up, pin 1 top-left. Power **red**, ground grey, clocks blue. Verify against the datasheet before powering up.

Legend — reading the pinouts *notation*

`/NAME` **Active LOW** — true / asserted when the pin is at logic **LOW** (~0 V). A leading **/** (datasheets often use an overline) marks these: e.g. `/RES /IRQ /CS /WE`.

`NAME` **Active HIGH / normal** — asserted when the pin is **HIGH** (~+5 V). No slash. `R/W` is a level: HIGH = read, LOW = write.

Colour key: Vcc / Vdd = power · **GND / Vss = ground** · **φ = clock**. **Orientation:** chip seen from the top, notch up — pin 1 top-left, numbers run **down the left** then **up the right**.

Z80A CPU *40-DIP*

A11	1	40	A10
A12	2	39	A9
A13	3	38	A8
A14	4	37	A7
A15	5	36	A6
CLK	6	35	A5
D4	7	34	A4
D3	8	33	A3
D5	9	32	A2
D6	10	31	A1
Vcc	11	30	A0
D2	12	29	GND
D7	13	28	/RFSH
D0	14	27	/M1
D1	15	26	/RESET
/INT	16	25	/BUSRQ
/NMI	17	24	/WAIT
/HALT	18	23	/BUSAK
/MREQ	19	22	/WR
/IORQ	20	21	/RD

Spectrum CPU @ 3.5 MHz. The ULA inserts wait states for contended RAM. **Fails:** dead machine, no boot.

ULA 5C/6C001 gate array *40-DIP*

/CAS	1	40	GND
/WR	2	39	Q 14MHz
/RD	3	38	/MREQ
/WE	4	37	A15
A0	5	36	A14
A1	6	35	/RAS
A2	7	34	/ROMCS
A3	8	33	/IOULA
A4	9	32	CLK
A5	10	31	D7
A6	11	30	D6
/INT	12	29	D5
+5V	13	28	SOUND
+5V	14	27	D4
U	15	26	T4
V	16	25	D3
/Y	17	24	T3
D0	18	23	T2
T0	19	22	D2
T1	20	21	D1

Ferranti ULA: video, RAM timing/contention, keyboard read, beeper, tape I/O. Pinout verified against the comp.sys.sinclair FAQ (pin 1 = /CAS, pin 40 = GND). Runs hot — a common failure. **Fails:** no/garbled display, no sound/tape.

4116 DRAM 16K×1 *16-DIP*

Vbb -5V	1	16	Vss
DIN	2	15	/CAS
/WE	3	14	DOUT
/RAS	4	13	A6
A0	5	12	A3
A2	6	11	A4
A1	7	10	A5
Vdd +12V	8	9	Vcc +5V

Lower 16K RAM = eight 4116 (needs +5/-5/+12V). Upper 32K = 4532/4164-class. **Fails:** crash/garbage in low RAM = bad 4116 (check the -5V/+12V rails).

2716/2316 ROM 2K×8 *24-DIP*

A7	1	24	Vcc
A6	2	23	A8
A5	3	22	A9
A4	4	21	Vpp
A3	5	20	/OE
A2	6	19	A10
A1	7	18	/CE
A0	8	17	O7
O0	9	16	O6
O1	10	15	O5
O2	11	14	O4
Vss	12	13	O3

16K BASIC ROM (usually one 23128-class mask ROM, shown here as the 2K 2716 family pinout). **Verify ROM size/pinout for your board.**

Memory & power *info*

CPU	Z80A @ 3.5 MHz
RAM	16K (4116) + 32K (4532)
ROM	16K mask
Rails	+5V, -5V, +12V (4116 needs all three)

Lose the -5V or +12V rail and the lower RAM (and often the ULA) misbehave.

Test & Repair *tips*

Hot ULA / no display	ULA (socket it)
No boot	Z80 / ROM / clock
Low-RAM garbage	4116 + -5/+12V rails
No beep/tape	ULA

Most Spectrum faults trace to the ULA, the 4116 lower RAM, or the multi-rail PSU.

Proofread checklist ⚠ verify these

BASIC ROM	exact part & pinout (16K vs the 2K family shown)
Upper RAM	4532 vs 4164 wiring

(Z80, ULA, 4116 pinouts now verified)

Amber pins and **⚠ flagged chips** are my best reconstruction from incomplete/garbled sources — confirm each against a datasheet before use.